

# Semantic Similarity in Latent Space for LLM Prediction Evaluation

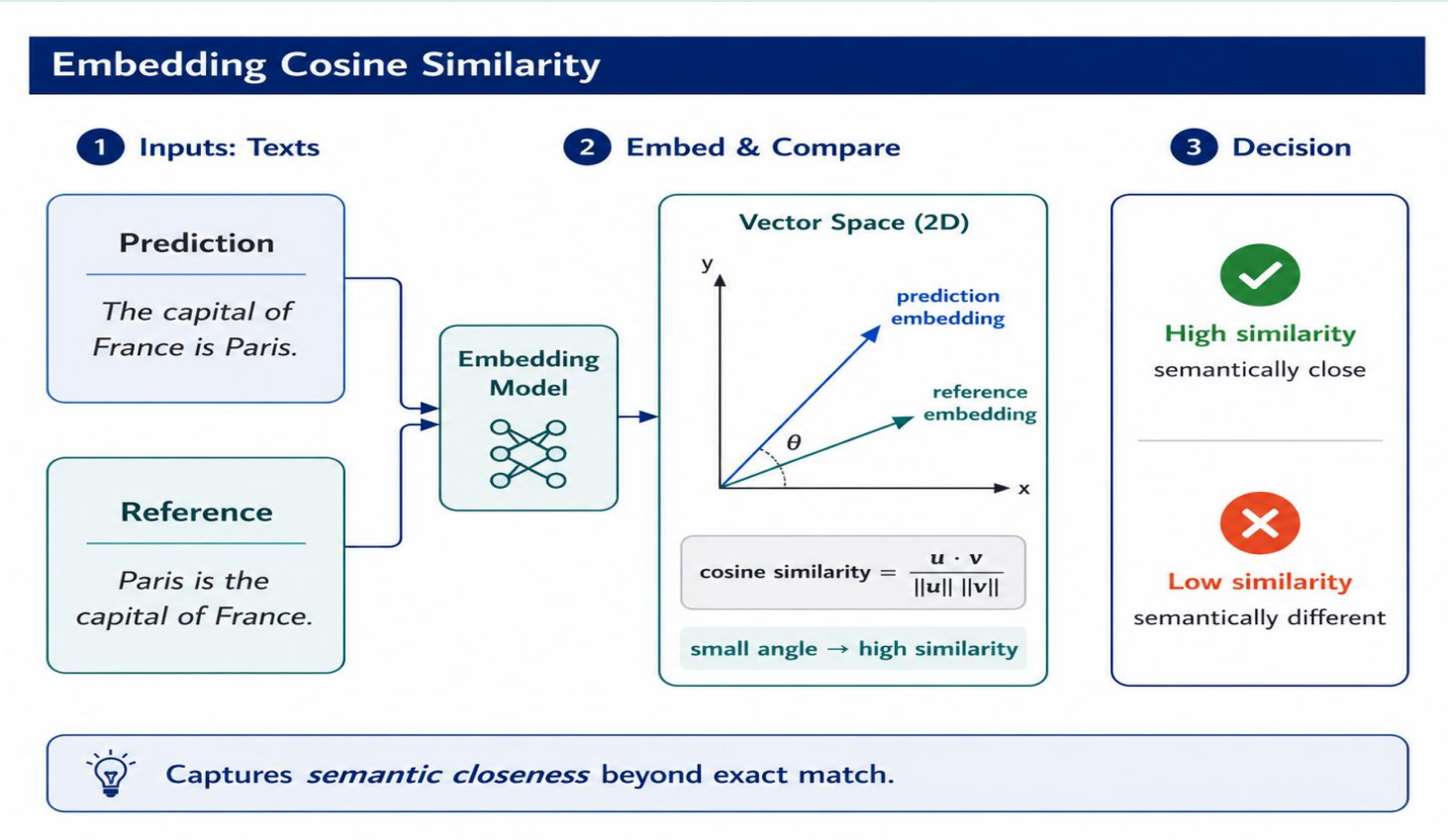
Team Members: Che FENG, YanZhe XIE, Ye GUO  
AIAA 4051 Final Research Project, Introduction to NLP, HKUST(GZ)

## Problem Background

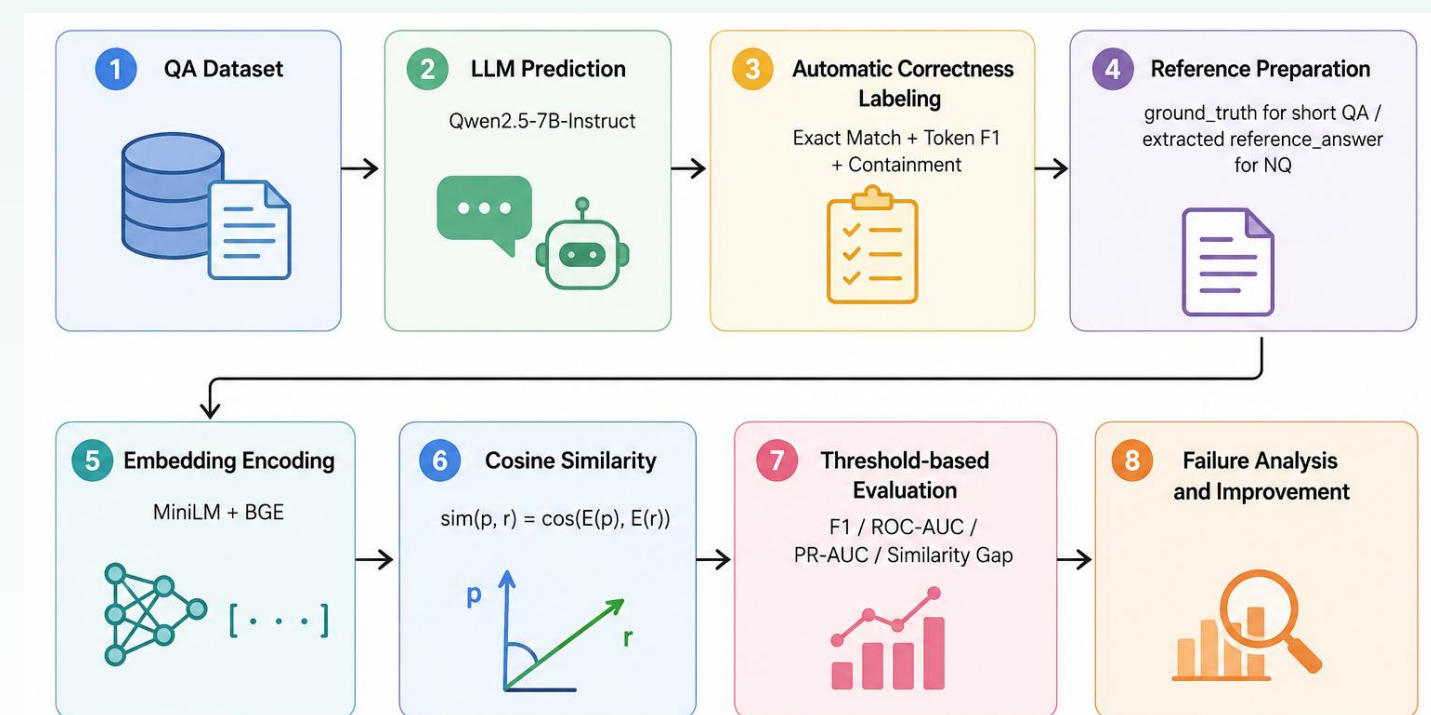
LLMs can produce fluent QA answers, but judging correctness is difficult. Exact-match metrics miss valid paraphrases, while embedding similarity may capture meaning better. However, semantic similarity can confuse topic overlap with factual correctness, especially in long answers.

### Research Question:

Can **cosine similarity** between prediction and reference embeddings distinguish correct from incorrect LLM-generated QA answers?



## Methodology



The correctness label is **automatically generated** rather than manually annotated. It combines exact match, token-level F1, and containment rules. Therefore, our evaluation measures whether embedding similarity agrees with a reproducible automatic correctness signal, while failure analysis identifies where this signal is noisy or incomplete.

## Experiment

### Experiment

We evaluate the similarity-based correctness signal on both **short-form** and **long-form** QA datasets. The main experimental variable is reference granularity: short-form datasets usually provide compact answer spans, while NQ originally provides long Wikipedia evidence passages.

### Setup

LLM: Qwen2.5-7B-Instruct

Embedding models: **MiniLM** and **BGE**

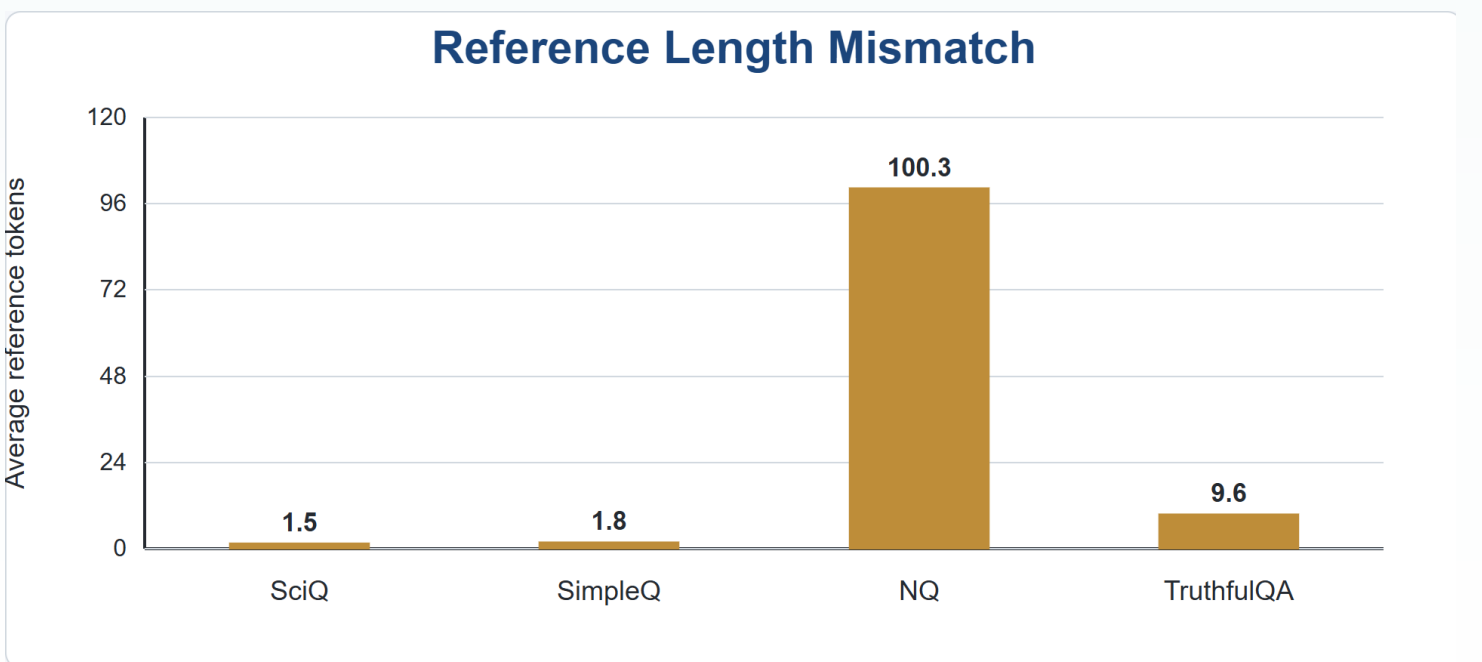
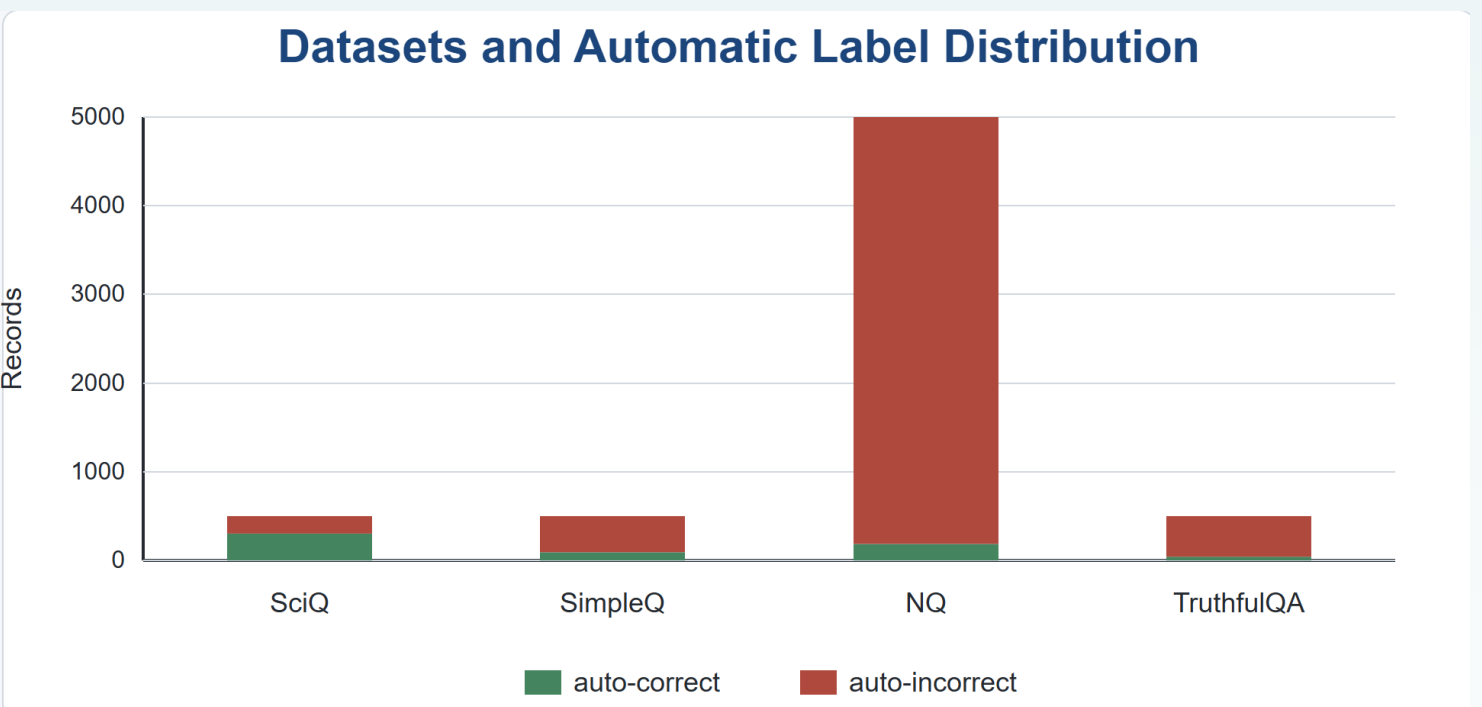
Similarity metric: cosine similarity

Decision rule: similarity  $\geq$  threshold  $\rightarrow$  predicted correct

Labels: rule-based automatic correct\_label

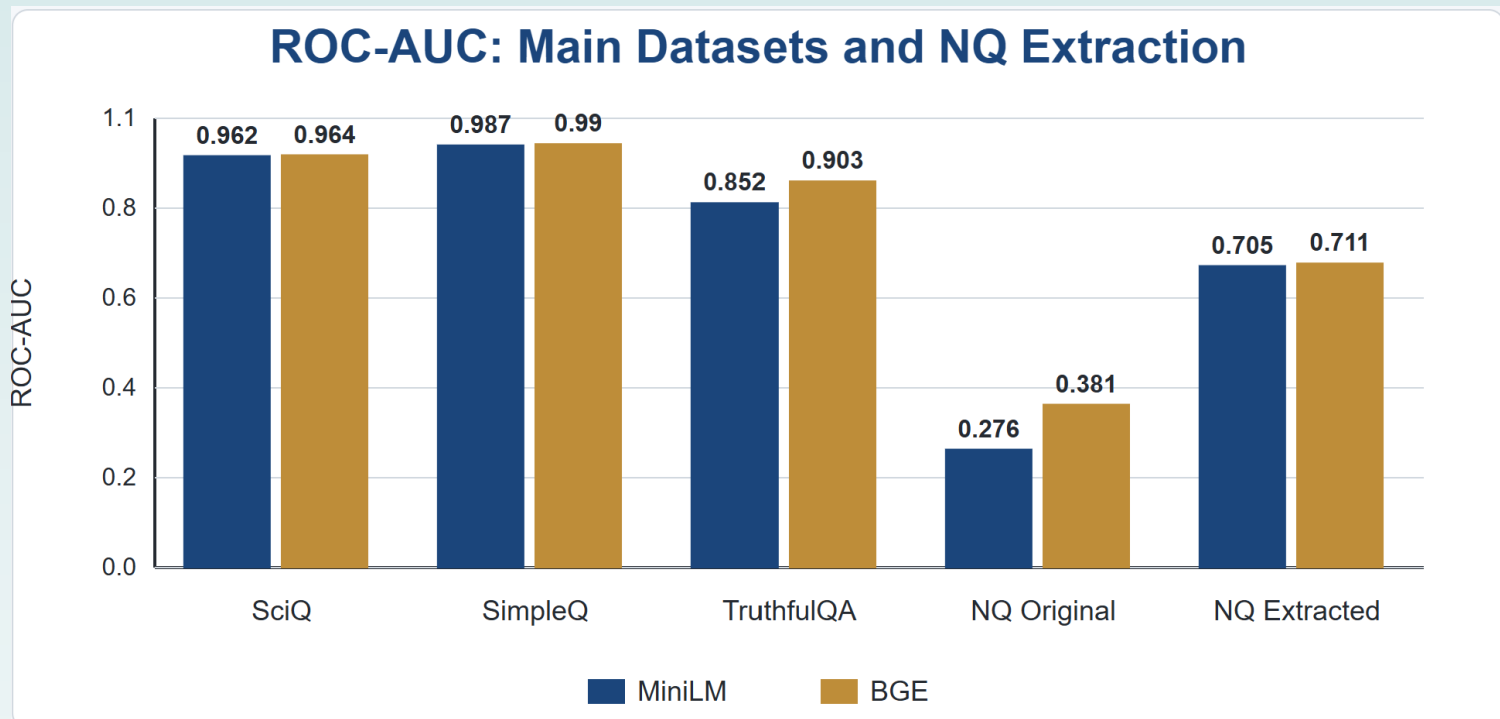
| Metric             | Purpose                                                       |
|--------------------|---------------------------------------------------------------|
| ROC-AUC            | Measures ranking quality across all thresholds                |
| Best F1            | Finds the strongest threshold-based classifier                |
| Fixed-threshold F1 | Tests a simple global threshold                               |
| Similarity Gap     | Difference between correct and incorrect mean similarity      |
| Failure Counts     | Counts high-similarity wrong and low-similarity correct cases |

### Dataset

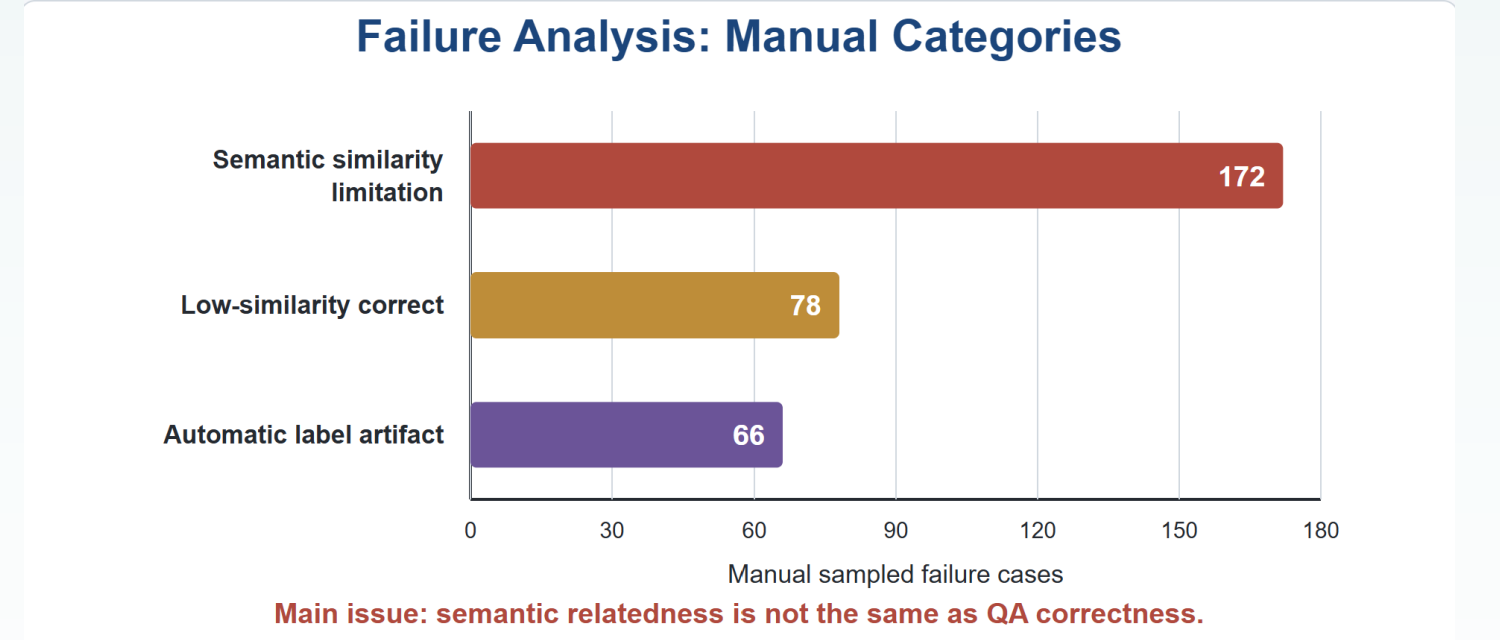


## Results

| Dataset      | Type       | MiniLM ROC-AUC | BGE ROC-AUC | Finding                             |
|--------------|------------|----------------|-------------|-------------------------------------|
| SciQ         | Short-form | 0.962          | 0.964       | Strong signal                       |
| SimpleQ-Wiki | Short-form | 0.987          | 0.990       | Strongest signal                    |
| TruthfulQA   | Long-form  | 0.852          | 0.903       | Useful but harder                   |
| NQ Original  | Long-form  | 0.276          | 0.381       | Fails under long-reference mismatch |



- Short-form QA shows **strong separation** between correct and incorrect predictions.
- TruthfulQA remains positive but is harder due to open-ended factual statements.
- Original NQ fails because short answers are compared with **long evidence passages**.



**Failure Analysis: What the Errors Mean**

|                                            |                                                                                 |                                              |                                                  |
|--------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------|
| <b>Topic overlap</b><br>same passage topic | <b>Granularity</b><br>too broad / specific                                      | <b>Label artifact</b><br>alias or paraphrase | <b>Context dilution</b><br>answer hidden in text |
| <b>High-sim wrong</b>                      | topical relatedness, long passages, or semantically close but incorrect answers |                                              |                                                  |
| <b>Low-sim correct</b>                     | short-answer vs long-context mismatch or extra generated context                |                                              |                                                  |
| <b>Label artifact</b>                      | automatic rules stricter than human judgment for aliases and paraphrases        |                                              |                                                  |

## Advanced

### 1. Baselines & Ablation Studies

We treat every improvement as an ablation. Starting from sentence-level embedding cosine, we compare lexical baselines, entity/token overlap, embedding-overlap hybrids, NQ reference extraction, prediction-span similarity, factual conflict penalties, and reduced multi-view hybrid scores.

#### Key results:

- NQ reference\_answer extraction improves ROC-AUC from **0.269/0.391** to **0.705/0.711**.
- Span-level similarity greatly improves ranking, with best F1 around **0.80**.
- Factual conflict penalties reduce high-similarity wrong cases caused by number/date/entity mismatch.
- Unit 6 span-guarded multi-view hybrid gives the best fixed-threshold performance: **MiniLM F1 = 0.725**, **BGE F1 = 0.701**.

### 2. Theory: Multi-View Embedding Evaluator

#### Core Idea

Correctness  $\neq$  semantic similarity only

**Correctness = semantic match + lexical evidence + factual-unit consistency + answer granularity + question-type awareness**

This keeps embedding similarity as one useful signal, but prevents it from becoming the only judge of factual correctness.

### 3. Case Studies

#### Case Study1

| Field              | Content                                                       |
|--------------------|---------------------------------------------------------------|
| Question           | Which river is the water source for Victoria Falls?           |
| Prediction         | Zambezi River                                                 |
| Original Reference | Long passage about Victoria Falls                             |
| Problem            | Correct short answer is compared with a long evidence passage |
| Fix                | Extract reference_answer = Zambezi River                      |

#### Case Study2

| Field       | Content                                   |
|-------------|-------------------------------------------|
| Question    | Who owns Aquarius Casino in Laughlin, NV? |
| Reference   | Golden Entertainment                      |
| Prediction  | Caesars Entertainment                     |
| Problem     | Same topic, wrong entity                  |
| Needed View | Entity consistency / verifier             |